makes the crystal lattice unstable. Being strongly basic, alkali metals except lithium form solid bicarbonates. No other metal forms solid bicarbonates.

$M_2CO_3 + CO_2 + H_2O \rightarrow 2\ MHCO_3$

(M = Na, K, Rb, Cs)

All the carbonates and bicarbonates are soluble in water and their solubilities increase rapidly on descending the group. This is due to the reason that lattice energies decrease more rapidly than their hydration energies on moving down the group.

5.3.1 Important compounds of alkali metals:

Sodium Carbonate $Na_2CO_3.10H_2O$ (Washing soda):

Sodium carbonate is one of the important inorganic compounds used in industries. It is prepared by Solvay process. In this process, ammonia is converted into ammonium carbonate which then converted to ammonium bicarbonate by passing excess carbon dioxide in a sodium chloride solution saturated with ammonia. The ammonium bicarbonate thus formed reacts with the sodium chloride to give sodium bicarbonate and ammonium chloride. As sodium bicarbonate has poor solubility, it gets precipitated. The sodium bicarbonate is isolated and is heated to give sodium carbonate. The equations involved in this process are,

$2NH_3 + H_2O + CO_2 \rightarrow (NH_4)_2CO_3$

$(NH_4)_2CO_3 + H_2O + CO_2 \rightarrow 2\ NH_4HCO_3$

$NH_4HCO_3 + NaCl \rightarrow NH_4Cl + NaHCO_3$

$2\ NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

The ammonia used in this process can be recovered by treating the resultant ammonium chloride solution with calcium hydroxide. Calcium chloride is formed as a by-product.

Properties:

Sodium carbonate, commonly known as washing soda, crystallises as decahydrate which is white in colour. It is soluble in water and forms an alkaline solution. Upon heating, it looses the water of crystallisation to form monohydrate. Above 373 K, the monohydrate becomes completely anhydrous and changes to a white powder called soda ash.

$Na_2CO_3 \cdot 10H_2O \rightarrow Na_2CO_3 \cdot H_2O + 9H_2O$

$Na_2CO_3 \cdot H_2O \rightarrow Na_2CO_3 + H_2O$

Uses:

i. Sodium carbonate known as washing soda and is used mainly for laundering.

ii. It is also used in water treatment to convert the hard water to soft water.

iii. It is used in the manufacturing of glass, paper, paint etc...

Sodium chloride NaCl (Cooking salt or Table salt):

Sodium chloride is isolated by evaporation from sea water which contains

11th Chemistry Chapter-5 Vol-1 EM.indd 133 09-01-2023 10:54:34